

FRACTIONS MADE FUN

Dividing Fractions

A step-by-step lesson for primary students on how to divide fractions using the **Keep–Change–Flip** method.

1. What Is a Fraction?

A fraction represents a **part of a whole**. When something is divided into equal parts, each part is a fraction.

Term	Meaning	Example
Numerator	Top number – how many parts we have	3 in $\frac{3}{4}$
Denominator	Bottom number – total equal parts	4 in $\frac{3}{4}$
Proper fraction	Numerator < Denominator	$\frac{2}{5}$
Improper fraction	Numerator > Denominator	$\frac{7}{4}$
Mixed number	Whole number + fraction	1 and $\frac{3}{4}$

2. What Does Dividing Fractions Mean?

Dividing fractions means finding out **how many times one fraction fits into another**.

Think of it like asking: "How many halves fit into three quarters?"

■ Real-life example

You have $\frac{3}{4}$ of a pizza. Each person gets $\frac{1}{2}$ a pizza's worth. How many people can you feed?
Answer: 1 and $\frac{1}{2}$ people!

3. The Five Steps: Keep–Change–Flip–Multiply–Simplify

1	KEEP	Keep the first fraction exactly as it is.	$\frac{3}{4} \div \frac{1}{2} \rightarrow$ Keep $\frac{3}{4}$
2	CHANGE	Change the division sign ($\div$) to a multiplication sign ($\times$).	$\frac{3}{4} \div \frac{1}{2} \rightarrow \frac{3}{4} \times$...
3	FLIP	Flip the second fraction (swap numerator and denominator). This is called finding the reciprocal.	$\frac{1}{2}$ flipped $\rightarrow \frac{2}{1}$

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MULTIPLY

Multiply straight across: top $\times$ top, bottom $\times$ bottom.

$$\frac{3}{4} \times \frac{2}{1} = \frac{6}{4}$$

$$(3 \times 2 = 6, 4 \times 1 = 4)$$

5

SIMPLIFY

Reduce to lowest terms by dividing by the greatest common factor (GCF).

$$\frac{6}{4} = \frac{3}{2} = 1 \text{ and } \frac{1}{2}$$

4. Worked Example: $\frac{2}{3} \div \frac{4}{5}$

Step	Action	Result
1 – Keep	Keep the first fraction	$\frac{2}{3}$
2 – Change	Change $\div$ to $\times$	$\frac{2}{3} \times$
3 – Flip	Flip $\frac{4}{5}$ to get its reciprocal	$\frac{5}{4}$
4 – Multiply	$\frac{2}{3} \times \frac{5}{4} = (2 \times 5) / (3 \times 4)$	$\frac{10}{12}$
5 – Simplify	Divide top & bottom by GCF (2)	$\frac{5}{6} \checkmark$

■ Final Answer: $\frac{2}{3} \div \frac{4}{5} = \frac{5}{6}$

5. Word Problems

Read each problem carefully, then use the Keep–Change–Flip method to solve it.

Q1. ■ Chocolate Bars

Answer: _____

Priya has $\frac{3}{4}$ of a chocolate bar. She wants to share it equally among **3 friends**. How much does each friend get?

Q2. ■ Recipe Ingredients

Answer: _____

Arjun needs $\frac{2}{3}$ **cup** of flour. His measuring cup holds $\frac{1}{6}$ **cup**. How many scoops does he need?

Q3. ■ Ribbon Pieces

Answer: _____

Meera has $\frac{4}{5}$ **metre** of ribbon. She cuts it into pieces that are $\frac{1}{10}$ **metre** each. How many pieces can she make?

6. Practice Problems

Solve these on your own using the steps you've learned!

Q1. $\frac{2}{3} \div \frac{1}{4} =$ _____

Q2. $\frac{5}{6} \div \frac{2}{3} =$ _____

Q3. $\frac{3}{5} \div \frac{1}{2} =$ _____

Q4. $\frac{4}{7} \div \frac{2}{5} =$ _____

Q5. $\frac{1}{2} \div \frac{3}{4} =$ _____

Q6. $\frac{7}{8} \div \frac{1}{4} =$ _____

7. Common Mistakes to Avoid

■ Forgetting to flip the second fraction

■ Always flip the **SECOND** fraction, not the first!

■ Multiplying the wrong fractions

■ After flipping, multiply top×top and bottom×bottom.

■ Not simplifying the final answer

■ Always check if you can reduce your fraction further.

■ Flipping the first fraction instead of the second

■ Only the divisor (second fraction) gets flipped.

8. Key Takeaways

KEEP	Keep the first fraction exactly as it is
CHANGE	Change the $\div$ sign to a $\times$ sign
FLIP	Flip the second fraction (find the reciprocal)
MULTIPLY	Multiply numerators together, then denominators
SIMPLIFY	Simplify to the lowest terms

Answer Key

Problem	Working	Answer
$2/3 \div 1/4$	$2/3 \times 4/1 = 8/3$	2 and $2/3$
$5/6 \div 2/3$	$5/6 \times 3/2 = 15/12$	1 and $1/4$
$3/5 \div 1/2$	$3/5 \times 2/1 = 6/5$	1 and $1/5$
$4/7 \div 2/5$	$4/7 \times 5/2 = 20/14$	1 and $3/7$
$1/2 \div 3/4$	$1/2 \times 4/3 = 4/6$	$2/3$
$7/8 \div 1/4$	$7/8 \times 4/1 = 28/8$	3 and $1/2$
Word Q1	$3/4 \div 3 = 3/4 \times 1/3 = 3/12$	$1/4$
Word Q2	$2/3 \div 1/6 = 2/3 \times 6/1 = 12/3$	4 scoops
Word Q3	$4/5 \div 1/10 = 4/5 \times 10/1 = 40/5$	8 pieces

■ You've got this! Keep practising and fractions will feel like second nature.

